Web of Duality of Bosonic and Fermionic CFT

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Abstract

This is a template of Typst note

Contents

1 Background Gauge Field Language

The bosonization of Ising and Majorana CFT are commonly written in a background gauge field language, here I'll give a nonrigorous brief introduction.

1.1 $\mathbb{Z}_2$ Gauge Field

1.1.1 $\mathbb{Z}_2$ Gauge Theory

In fact, boundary conditions in the bosonic theory can be understood in a unified way using the language of a $\mathbb{Z}_2$ gauge theory. We introduce a background gauge field

$$A \in H^1\left(\Sigma_g, \mathbb{Z}_2\right) \tag{1}$$

This object admits several equivalent interpretations:

- as a cohomology class,
- as a flat $\mathbb{Z}_2$ connection,
- as a homomorphism from the fundamental group $A \in \mathrm{Hom}\left(\pi_1\left(\Sigma_g\right), \mathbb{Z}_2\right)$.

Concretely, this means that A assigns a $\mathbb{Z}_2$ value to each non-contractible cycle. Choosing a canonical basis $\left\{a_1, b_1, ..., a_g, b_g\right\}$, we write

$$A(a_i) = \alpha_i \in \mathbb{Z}_2, \quad A(b_i) = \beta_i \in \mathbb{Z}_2. \tag{2}$$

Thus the gauge field is completely specified by

$$A = \left(\alpha_1, \beta_1, ..., \alpha_g, \beta_g\right) \in (\mathbb{Z}_2)^{2g}. \tag{3}$$

1.1.2 Cup Product

Given two $\mathbb{Z}_2$ gauge fields $A, B \in H^1\left(\Sigma_g, \mathbb{Z}_2\right)$, their cup product is an element

$$A \cup B \in H^2\left(\Sigma_g, \mathbb{Z}_2\right) \approx \mathbb{Z}_2. \tag{4}$$

On a closed Riemann surface, this evaluates to a number:

$$\int_{\Sigma_g} A \cup B \in \mathbb{Z}_2. \tag{5}$$

In terms of the cycle basis, if $A = (\alpha_i, \beta_i)$ and $B = (\alpha_i', \beta_i')$, then

$$\int_{\Sigma_g} A \cup B = \sum_{i=1}^{g} (\alpha_i \beta_i' + \beta_i \alpha_i') \mod 2. \tag{6}$$

Since we are working in $\mathbb{Z}_2$, this simplifies to

$$\int_{\Sigma_g} A \cup B = \sum_{i=1}^{g} (\alpha_i \beta_i' + \beta_i \alpha_i') \mod 2. \tag{7}$$

For example for a torus gauge field we have the following different results:

$\int_{T^2} A \cup B$	$A_0 = (0,0)$	$A_1 = (1,0)$	$A_2 = (0,1)$	$A_3 = (1,1)$
$A_0 = (0,0)$	0	0	0	0
$A_1 = (1,0)$	0	0	1	1
$A_2 = (0,1)$	0	1	0	1
$A_3 = (1,1)$	0	1	1	0

Table 1: Cup product integral on the torus T^2 for the four $\mathbb{Z}_2$ gauge fields. The entries are computed modulo 2.

1.1.3 Bosonic Partition Function Coupled to Background Field

We can understand partition functions with different boundary conditions of bosonic theory (or more generally, a $\mathbb{Z}_2$ defect twisted theory with different ways of twisting) as coupling to different background $\mathbb{Z}_2$ gauge fields. Namely, we write

$$Z[A], \quad A \in H^1(\Sigma_g, \mathbb{Z}_2), \tag{8}$$

where A specifies the twisting of fields along non-contractible cycles. We commonly use:
- 0 for Periodic Boundary Condition (didn't cross a defect)
- 1 for Anti-Periodic Boundary Condition (crossing a defect)

For example, on the torus $\Sigma_1 = T^2$, we choose cycles (a, b) and write $A = (\alpha, \beta) \in \mathbb{Z}_2 \times \mathbb{Z}_2$. Then the boundary conditions are:

$$Z_{\mathrm{P,P}} = Z[(0,0)], \quad Z_{\mathrm{P,AP}} = Z[(0,1)], \quad Z_{\mathrm{AP,P}} = Z[(1,0)], \quad Z_{\mathrm{AP,AP}} = Z[(1,1)]. \tag{9}$$

"In fact I don't understand why coupling to a gauge field is equal to inserting a Symmetry defect line, nevertheless, the calculational result is the same in partition functions"

1.2 Spin Structure

1.2.1 Spin Structure

Mathematically, we can define a spin structure using the quadratic refinement of a $\mathbb{Z}_2$ gauge field:

$$\rho : H^1(\Sigma_g, \mathbb{Z}_2) \to \mathbb{Z}_2. \tag{10}$$

But physically, we can just understand it as specifying the boundary conditions of fermions along the non-contractible cycles of the surface. Just as for the $\mathbb{Z}_2$ gauge field, we can label a number for each non-contractible cycle, with:
- 1 for Periodic Boundary Condition (Ramond)
- 0 for Anti-Periodic Boundary Condition (Neveu-Schwarz)

> **Remark**
>
> This is opposite from the bosonic counter part, when writing a twisted bosonic theory in terms of coupling to background gauge field. Back then we use 0 for periodic.

For example, on a torus T^2 with a- and b-cycles, a spin structure determines whether the fermion is periodic (Ramond) or antiperiodic (Neveu-Schwarz) along each cycle:

$$\rho = (\text{a-cycle}, \text{b-cycle}) \in \{\mathrm{AP}, \mathrm{P}\}^2. \tag{11}$$

Furthermore, two spin structures can be related by a background $\mathbb{Z}_2$ gauge field $A \in H^1(\Sigma_g, \mathbb{Z}_2)$ via

$$\rho + A = (\rho_a + A_a \bmod 2, \quad \rho_b + A_b \bmod 2), \tag{12}$$

which physically corresponds to flipping the fermion boundary condition along cycles where A is nonzero.

1.2.2 Arf Invariant

The Arf invariant is a $\mathbb{Z}_2$-valued number that depends only on the spin structure of a surface. Physically, it distinguishes between "even" and "odd" spin structures. Mathematically, it can be written using the language of quadratic refinements acting on basis.

On the torus T^2, with standard cycles (a, b), we can label spin structures by numbers along each cycle (remember):

$$0 \text{ for Anti-Periodic (AP)}, \quad 1 \text{ for Periodic (P)}. \tag{13}$$

Then the four spin structures and their Arf invariants are:

Spin structure (ρ_a, ρ_b)	a-cycle	b-cycle	$\mathrm{Arf}(\rho)$
$(0, 0)$	0	0	0
$(1, 0)$	1	0	0
$(0, 1)$	0	1	0
$(1, 1)$	1	1	1

1.3 Useful Results of Arf

- **Result 1:** For sum over cup products we have:

$$\frac{1}{2^g} \sum_s (-1)^{\int s \cup t} = \begin{cases} 2^g & \text{if } \ t = 0 \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

- **Result 2:** For Arf invariant of spin structure act on by multi gauge fields we have:

$$\mathrm{Arf}[a + b + \rho] = \mathrm{Arf}[a + \rho] + \mathrm{Arf}[b + \rho] + \mathrm{Arf}[\rho] + \int a \cup b \tag{15}$$

- **Result 3:** sum over many:

$$\frac{1}{2^g} \sum_s (-1)^{\mathrm{Arf}(s+\rho) + \ \mathrm{Arf}[\rho] + \int s \cup t} = (-1)^{\mathrm{Arf}[t+\rho]} \tag{16}$$

1.4 Useful Partition Function Data

Finally, we can explicitly write out the partition functions with different spin structure (fermionic) or coupling to different background gauge field (Bosonic).

Partition Function	Spin Structure	Characters
$Z_{\mathrm{AP,AP}}$	$Z[(0, 0)]$	$\lvert\chi_0\rvert^2 + \lvert\chi_{\frac{1}{2}}\rvert^2 + \lvert(\chi)_0 \chi_{\frac{1}{2}} + \lvert(\chi)_{\frac{1}{2}} \chi_0$
$Z_{\mathrm{AP,P}}$	$Z[(0, 1)]$	$\lvert\chi_0\rvert^2 + \lvert\chi_{\frac{1}{2}}\rvert^2 - \lvert(\chi)_0 \chi_{\frac{1}{2}} - \lvert(\chi)_{\frac{1}{2}} \chi_0$
$Z_{\mathrm{P,AP}}$	$Z[(1, 0)]$	$2\ \lvert\chi_{\frac{1}{16}}\rvert^2$
$Z_{\mathrm{P,P}}$	$Z[(1, 1)]$	0

Table 2: Partition function and corresponding characters for Majorana CFT (P/AP).

Partition Function	BG Gauge Field	Characters
$Z_{\mathrm{P,P}}$	$Z[(0,0)]$	$\lvert\chi_0\rvert^2 + \lvert\chi_{\frac{1}{2}}\rvert^2 + \lvert\chi_{\frac{1}{16}}\rvert^2$
$Z_{\mathrm{P,AP}}$	$Z[(0,1)]$	$\lvert\chi_0\rvert^2 + \lvert\chi_{\frac{1}{2}}\rvert^2 - \lvert\chi_{\frac{1}{16}}\rvert^2$
$Z_{\mathrm{AP,P}}$	$Z[(1,0)]$	$\lvert(\chi)_0\chi_{\frac{1}{2}} + \lvert(\chi)_{\frac{1}{2}}\chi_0 + \lvert\chi_{\frac{1}{16}}\rvert^2$
$Z_{\mathrm{AP,AP}}$	$Z[(1,1)]$	$-\lvert(\chi)_0\chi_{\frac{1}{2}} - \lvert(\chi)_{\frac{1}{2}}\chi_0 + \lvert\chi_{\frac{1}{16}}\rvert^2$

Table 3: Partition function and corresponding characters for Ising CFT (P/AP).

Bibliography